5.1.2 Basic characteristics of fourth generation of drones

The fourth generation of drones embodies a revolutionary shift toward autonomous intelligence, real-time adaptability, and networked collaboration, integrating cutting-edge technologies that enable drones to function as independent, cognitive systems rather than remote-controlled machines. The following are the defining characteristics of this generation:

- *Artificial intelligence (AI) and machine learning (ML) integration*
  Fourth-generation drones utilize AI-driven algorithms and machine learning models for autonomous decision-making, pattern recognition, and adaptive mission execution. These systems continuously learn from operational data to improve flight precision, threat response, and mission efficiency without human oversight.
- *Swarm intelligence and cooperative autonomy*
  One of the most distinctive features of this generation is the ability to perform swarm-based operations, where multiple drones communicate and collaborate in real time to execute missions collectively. This distributed intelligence allows for self-organization, task-sharing, and dynamic adaptability in combat, reconnaissance, or disaster-relief missions.
- *Multi-domain interoperability*
  Fourth-generation drones are designed for multi-domain integration, capable of coordinating with manned aircraft, naval vessels, ground vehicles, and satellite systems. This interoperability enhances situational awareness and supports network-centric warfare and cross-domain operations through seamless data sharing.
- *Real-time data processing and edge computing*
  Equipped with onboard edge computing systems, these drones can process vast amounts of sensor and video data locally, reducing latency